

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Architecture**

Code No: **CSC165** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain the working of Booth's multiplication algorithm and perform multiplication of 50 and (-13) using the same algorithm.
2. Explain the instruction format of basic computer. Write down symbolic microinstruction for FETCH routine.
3. What is DMA? Explain the DMA controller with block diagram. How the DMA interact with I/O device.

Section B

Attempt any Eight questions[8×5=40]

4. What is arithmetic overflow? How it can be detected?
5. What do you mean by error correction codes? Explain any one of error correction technique.
6. Explain block diagram of microprogram sequencer in brief.
7. What is cache memory? Explain the elements of cache design.
8. What is micro-operation? Explain different arithmetic microorganisms.
9. What do you mean by pipelining concept? Discuss various pipeline hazards and their solutions in detail.
10. Write the program for following statement by using three, single, zero address instructions.
$$X = (A + B * C - D) / (E * F + G)$$
11. Differentiate between RISC and CISC architecture.
12. Write short notes on (Any TWO)a. Control ROMb. Common Bus Systemc. Flynn's Classification